

# Circuit Documentation

## Summary

This circuit is designed trimmer potentiometer is used to simulate a turbidity/sludge sensor during circuit testing and simulation . It utilizes an ESP32-S3 microcontroller to process the sensor data and control two LEDs and a piezo speaker. The green LED indicates good water quality, while the red LED and the piezo speaker provide a warning when the water quality is poor.

## Component List

1. **ESP32-S3**
  - **Description:** A powerful microcontroller with Wi-Fi and Bluetooth capabilities.
  - **Pins:** 3V3, RST, 4, 5, 6, 7, 15, 16, 17, 18, 8, 3, 46, 9, 10, 11, 12, 13, 14, 5V, GND, 19, 20, 21, 47, 48, 45, 0, 35, 36, 37, 38, 39, 40, 41, 42, 2, 1, RX, TX
2. **LED Two Pin (Red)**
  - **Description:** A standard red LED used for indicating warnings.
  - **Pins:** Cathode, Anode
3. **LED Two Pin (Green)**
  - **Description:** A standard green LED used for indicating normal conditions.
  - **Pins:** Cathode, Anode
4. **Trimmer Potentiometer**
  - **Description:** A variable resistor used as a sensor to measure water quality.
  - **Pins:** leg1, wiper, leg2
  - **Properties:** Resistance: 10,000 Ohms
5. **Piezo Speaker**
  - **Description:** A small speaker used to emit sound warnings.
  - **Pins:** pin1, pin2
6. **Resistor (220 Ohms)**
  - **Description:** A resistor used to limit current to the LEDs.
  - **Pins:** pin1, pin2
7. **Resistor (220 Ohms) (1)**
  - **Description:** Another resistor used to limit current to the LEDs.
  - **Pins:** pin1, pin2

## Wiring Details

### ESP32-S3

- **3V3:** Connected to leg1 of the Trimmer Potentiometer.
- **GND:** Connected to leg2 of the Trimmer Potentiometer, Cathode of both LEDs, and pin2 of the Piezo Speaker.
- **Pin 4:** Connected to the wiper of the Trimmer Potentiometer.

- **Pin 18:** Connected to pin1 of the Resistor (220 Ohms) (1), which is connected to the Anode of the Green LED.
- **Pin 19:** Connected to pin1 of the Resistor (220 Ohms), which is connected to the Anode of the Red LED.
- **Pin 14:** Connected to pin1 of the Piezo Speaker.

## LED Two Pin (Red)

- **Anode:** Connected to pin2 of the Resistor (220 Ohms).
- **Cathode:** Connected to GND of the ESP32-S3.

## LED Two Pin (Green)

- **Anode:** Connected to pin2 of the Resistor (220 Ohms) (1).
- **Cathode:** Connected to GND of the ESP32-S3.

## Trimmer Potentiometer

- **leg1:** Connected to 3V3 of the ESP32-S3.
- **wiper:** Connected to Pin 4 of the ESP32-S3.
- **leg2:** Connected to GND of the ESP32-S3.

## Piezo Speaker

- **pin1:** Connected to Pin 14 of the ESP32-S3.
- **pin2:** Connected to GND of the ESP32-S3.

## Resistor (220 Ohms)

- **pin1:** Connected to Pin 19 of the ESP32-S3.
- **pin2:** Connected to the Anode of the Red LED.

## Resistor (220 Ohms) (1)

- **pin1:** Connected to Pin 18 of the ESP32-S3.
- **pin2:** Connected to the Anode of the Green LED.

## Code Documentation

```
#define SENSOR_PIN 4
#define GREEN_LED 18
#define RED_LED 19
#define BUZZER 14
```

```

void setup() {
    Serial.begin(115200);

    pinMode(GREEN_LED, OUTPUT);
    pinMode(RED_LED, OUTPUT);
    pinMode(BUZZER, OUTPUT);

    digitalWrite(GREEN_LED, LOW);
    digitalWrite(RED_LED, LOW);
    digitalWrite(BUZZER, LOW);
}

void loop() {
    int sensorValue = analogRead(SENSOR_PIN);

    Serial.print("Sensor Value: ");
    Serial.println(sensorValue);

    if (sensorValue > 2000) {
        // High sludge level
        digitalWrite(RED_LED, HIGH);
        digitalWrite(GREEN_LED, LOW);
        digitalWrite(BUZZER, HIGH);

        Serial.println("WARNING: Tank Cleaning Required");
    } else {
        // Normal water quality
        digitalWrite(RED_LED, LOW);
        digitalWrite(GREEN_LED, HIGH);
        digitalWrite(BUZZER, LOW);

        Serial.println("Water Quality Good");
    }

    delay(500);
}

```

## Code Explanation

- **Setup Function:** Initializes the serial communication and sets the pin modes for the LEDs and buzzer. All outputs are initially set to LOW.
- **Loop Function:** Continuously reads the sensor value from the trimmer potentiometer. If the sensor value exceeds 2000, it indicates a high sludge level, turning on the red LED and buzzer while turning off the green LED. Otherwise, it indicates good water quality, turning on the green LED and turning off the red LED and buzzer. The sensor value is printed to the serial monitor every 500 milliseconds.